SOC Threat Detection, Alert Correlation & Incident Response

Cyber Security Internship - Week 11

Wazuh + Snort | Lab Investigation Report

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Program: DG Interns Hub - Cyber Security Internship

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Incident reference: DG-SOC-001

All testing in this report was done inside my own isolated VirtualBox lab on machines I own. No external systems were touched.

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1. Lab Environment and Time Conventions

Machine	OS / Role	IP address	Notes
Windows host (PowerShell)	Attacker / test machine	192.168.56.1	Generated ping, nmap scan and SSH logins
soc-snort (VirtualBox)	Ubuntu 24.04 LTS, Snort IDS + Wazuh agent (target)	192.168.56.102 (enp0s8, host-only)	Also has NAT interface 10.0.2.15 (not used for the tests). Wazuh shows this agent as **snort-agent**
wazuh-server (VirtualBox)	Wazuh 4.14.9 OVA (manager, indexer, dashboard)	192.168.56.103	Dashboard at https://192.168.56.103

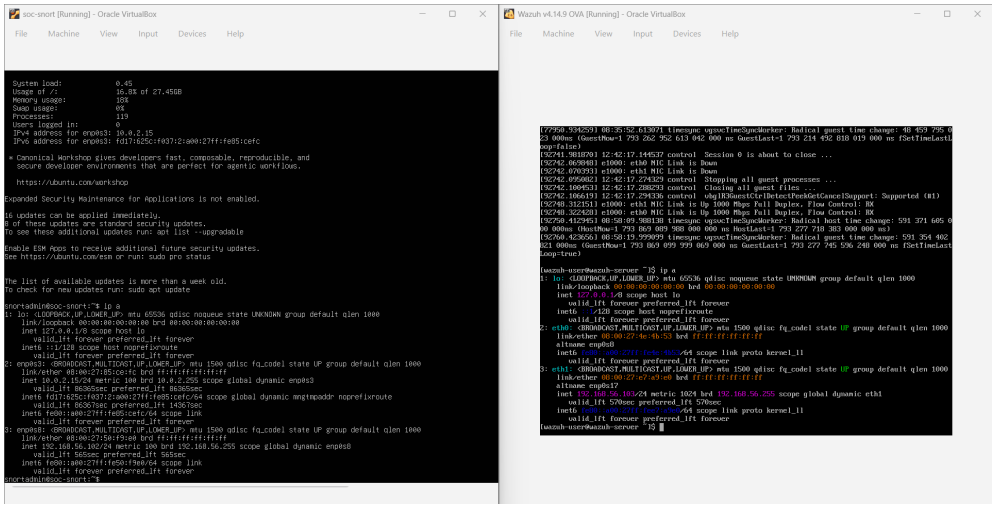


Figure 1: Lab machines - soc-snort (left) and Wazuh server (right) with their IP addresses.

Time conventions

- Linux logs (auth.log, Snort) are in **UTC**. Pakistan time (PKT) is UTC+5. The Wazuh dashboard shows PKT.

- Example: 09:10:59 UTC in auth.log = 14:10:59 PKT.
- The Wazuh alert timestamps are about 22-23 seconds earlier than the matching auth.log lines (for example, rule 40112 at 14:10:36 PKT vs the Accepted login at 09:10:59 UTC). This looks like clock drift on the Wazuh VM after it was resumed from a saved state, and it is noted wherever times are compared.
- Older entries dated 28 October 2026 in auth.log are from earlier Week 9/10 tests and are **not** counted in this incident.

2. SOC Alert Correlation (Notes)

2.1 What is security alert correlation?

Alert correlation means linking separate alerts and log events (from different tools, hosts or times) to see if they are part of one story. One failed login alone means little. Ten failed logins from one IP, then a successful login for the same account, is a pattern. Correlation turns many small signals into one meaningful finding. In this lab, Wazuh did this automatically with rule 40112 (multiple authentication failures followed by a success).

2.2 Why SOC analysts correlate multiple alerts

- **Reduce noise:** hundreds of low alerts become one incident to review.
- **See the full attack chain:** scan, then guess passwords, then log in, then act.
- **Raise confidence:** the same source IP in a network tool (Snort) and a host tool (Wazuh) is stronger evidence than either alone.
- **Prioritise:** correlated alerts get higher severity, so analysts work on the real risks first.
- **Cover blind spots:** each tool sees different things (see Section 5).

2.3 False Positive vs True Positive

Term	Meaning	Example
True Positive (TP)	The alert fired and the activity is really malicious or policy-breaking.	Brute-force from an unknown IP, followed by a login.
False Positive (FP)	The alert fired but the activity is harmless.	An admin mistyped a password twice; a monitoring tool pings the server every minute.
True Negative	No alert and nothing bad happened.	Normal logins, no alert.
False Negative	No alert but something bad happened (most dangerous).	Attack that no rule detects.

2.4 IOC (Indicator of Compromise)

An IOC is a piece of evidence that suggests a system was attacked or compromised. Examples: a suspicious IP address, a username being guessed, a file path, a domain, a port, a file hash, or a rule/signature ID. IOCs are used to search other logs, block attackers and share intelligence. The IOCs from this lab are listed in Section 8.

2.5 Event vs Alert vs Incident

Term	Meaning	Example from this lab
Event	Any recorded activity in a system. Most events are normal.	A line in auth.log: Failed password for wronguser.
Alert	An event (or group of events) that matched a detection rule and needs attention.	Wazuh rule 5710 or Snort SID 1:1000001 firing.
Incident	One or more alerts confirmed as a real security issue that needs a response.	DG-SOC-001: failed SSH logins followed by a successful login.

2.6 Threat Detection vs Incident Response

Threat Detection	Incident Response
Finding suspicious activity using tools (IDS, SIEM, logs).	Acting on a confirmed threat to limit damage and recover.

Threat Detection	Incident Response
Question: *Is something wrong?*	Question: *What do we do about it?*
Tools: Snort, Wazuh rules, log monitoring.	Steps: contain, eradicate, recover, learn.
Happens continuously (24/7 monitoring).	Starts when an incident is declared.

2.7 Role of L1, L2 and L3 analysts

Level	Main role	Typical tasks
L1 (Triage)	First line. Watches the dashboard and queue.	Review new alerts, filter false positives, collect basic evidence, open a ticket, escalate if real.
L2 (Investigation)	Deeper analysis of escalated alerts.	Correlate logs across tools, confirm True Positive, decide severity, contain (block IP, disable account), write findings.
L3 (Expert / Hunter)	Advanced threats and improvement.	Threat hunting, malware/forensic analysis, tune and write detection rules, lead major incidents, improve playbooks.

3. Wazuh Monitoring

Wazuh 4.14.9 (manager, indexer and dashboard) from Week 10 was used. The Wazuh agent runs on soc-snort and appears in the dashboard as **snort-agent**. After the Wazuh VM was resumed (it had been paused for about 10 days) the three services were restarted and the dashboard became reachable again.

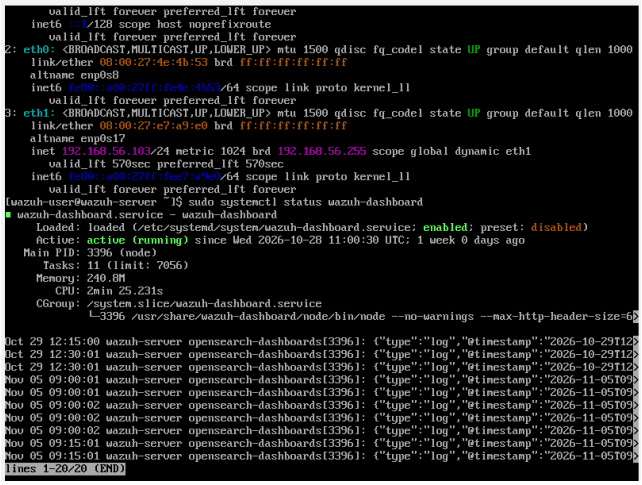


Figure 2: wazuh-dashboard service active (running) on the Wazuh server.

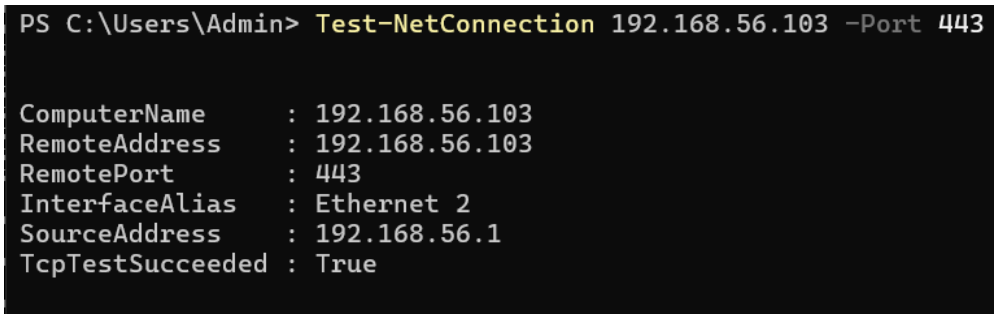


Figure 3: Port 443 on 192.168.56.103 reachable from the Windows host (TcpTestSucceeded: True).

3.1 Agent status

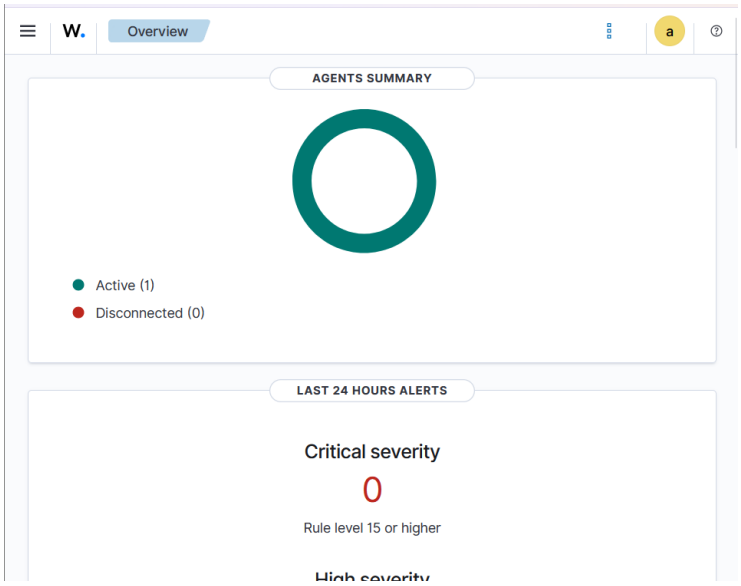


Figure 4: Agents summary - Active (1), Disconnected (0). Last 24 hours: Critical severity (level 15+) = 0.

3.2 Overall counters (last 24 hours)

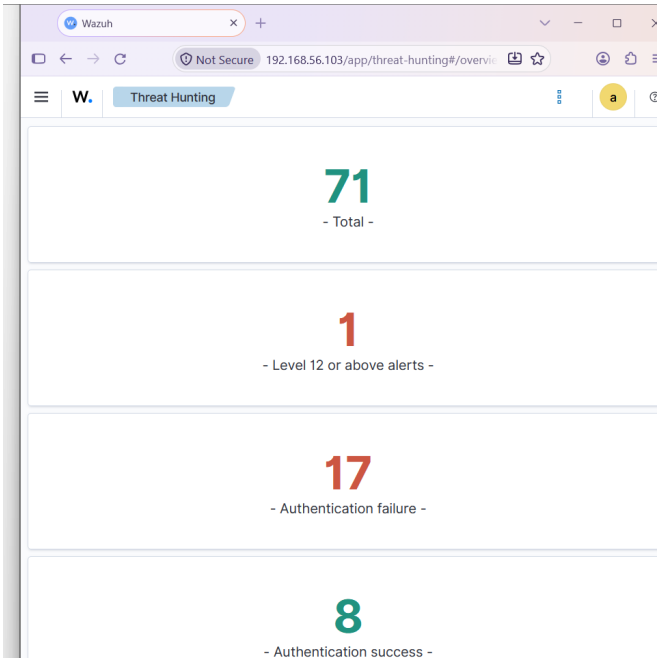


Figure 5: Threat Hunting summary - 71 total alerts, 1 alert at level 12 or above, 17 authentication failures, 8 authentication successes.

3.3 Failed login events from the test

Filtering the Events tab by the SSH authentication rule IDs shows the failed logins generated from the Windows PowerShell test.

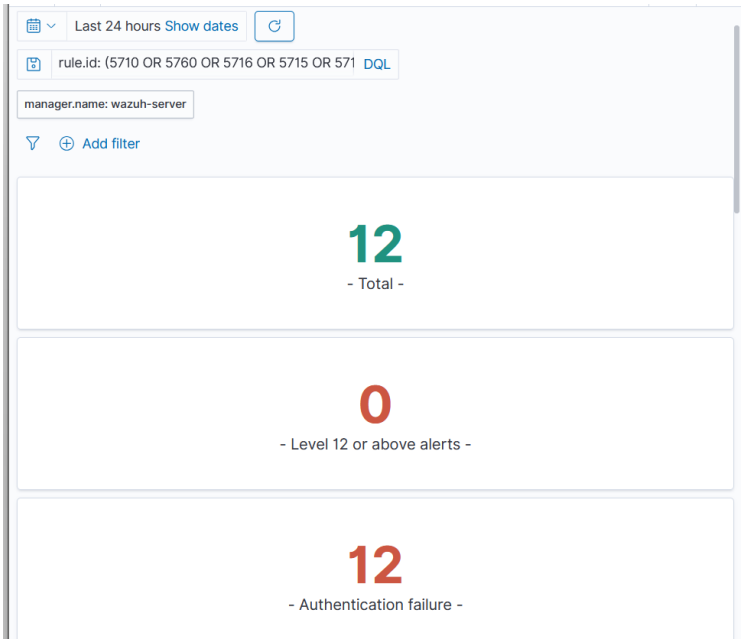


Figure 6: Filter rule.id (5710 OR 5760 OR 5716 OR 5715 OR 5712 OR 5763) - 12 events, all authentication failures, 0 at level 12+.

11 hits

Nov 4, 2026 @ 14:43:03.800 - Nov 5, 2026 @ 14:43:03.800

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1 fields sorted | Full screen

stamp	agent.name	rule.des...	rule.level	rule.id
2026 @ 14:07:34.178	snort-agent	sshd: auth...	5	5760
2026 @ 14:07:30.100	snort-agent	sshd: auth...	5	5760
2026 @ 14:07:17.926	snort-agent	sshd: Atte...	5	5710
2026 @ 14:07:17.926	snort-agent	sshd: Atte...	5	5710
2026 @ 14:07:07.012	snort-agent	sshd: Atte...	5	5710
2026 @ 14:07:02.501	snort-agent	sshd: Atte...	5	5710
2026 @ 14:06:56.445	snort-agent	sshd: Atte...	5	5710
2026 @ 14:06:54.438	snort-agent	sshd: Atte...	5	5710
2026 @ 14:06:50.396	snort-agent	sshd: Atte...	5	5710
2026 @ 14:06:38.642	snort-agent	sshd: Atte...	5	5710
2026 @ 14:06:24.591	snort-agent	sshd: Atte...	5	5710

Figure 7: Events table, 11 hits from agent snort-agent: rule 5760 (x2) and rule 5710 (x9), level 5, between 14:06:24 and 14:07:34 PKT.

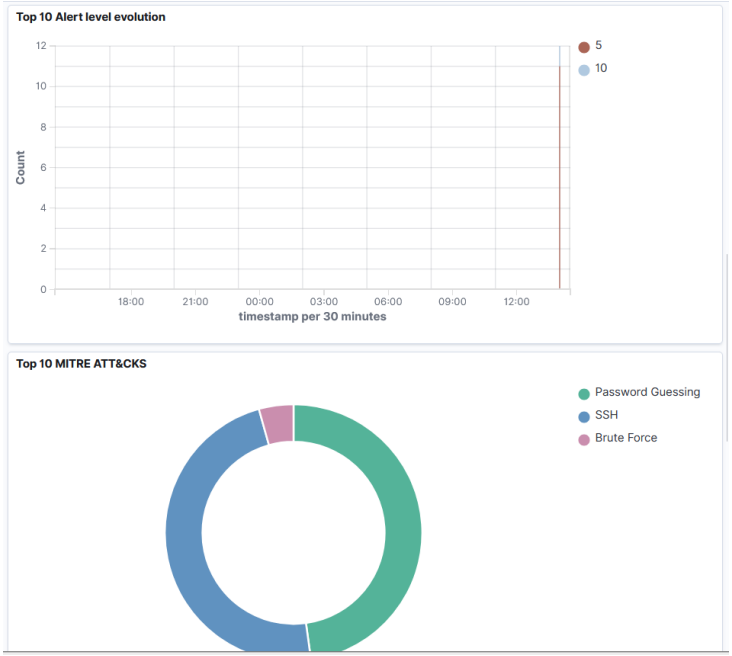


Figure 8: Alert level evolution (levels 5 and 10) and Top 10 MITRE ATT&CK: Password Guessing, SSH, Brute Force.

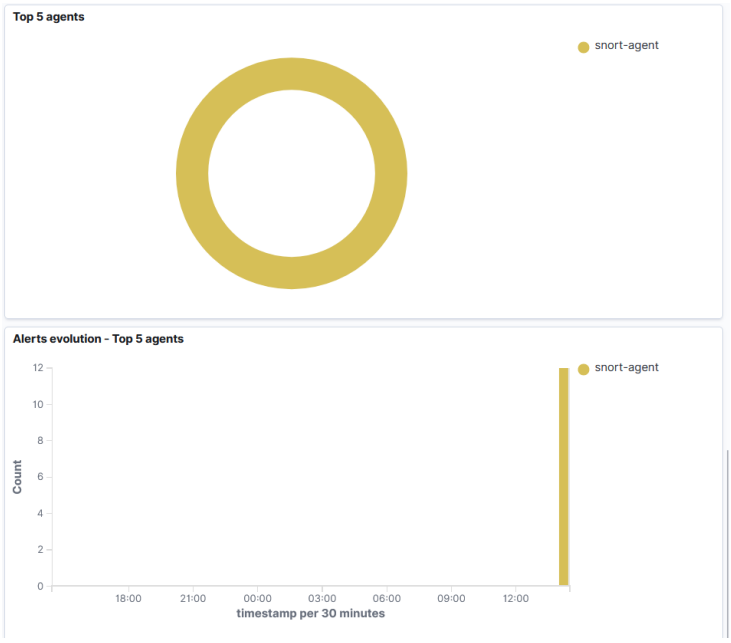


Figure 9: All 12 alerts came from one agent, snort-agent.

3.4 Correlation alert (level 12)

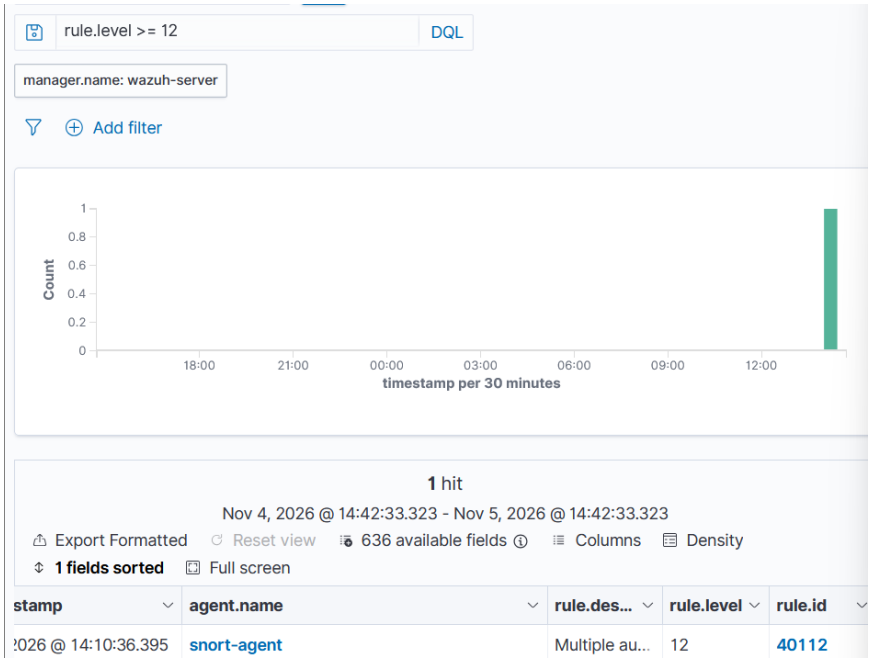


Figure 10: Query rule.level >= 12 returns 1 hit: rule 40112, level 12, agent snort-agent, Nov 5 2026 14:10:36 PKT.



Figure 11: The level 12 alert maps to MITRE ATT&CK Brute Force and Valid Accounts.

3.5 Successful login and file change searches

Rule 5715 (SSH authentication success) returned no result. This is expected here because Wazuh raised the correlation rule 40112 (failures followed by a success) for the successful login instead. The Wazuh counter shows 8 authentication-success events in the last 24 hours overall.

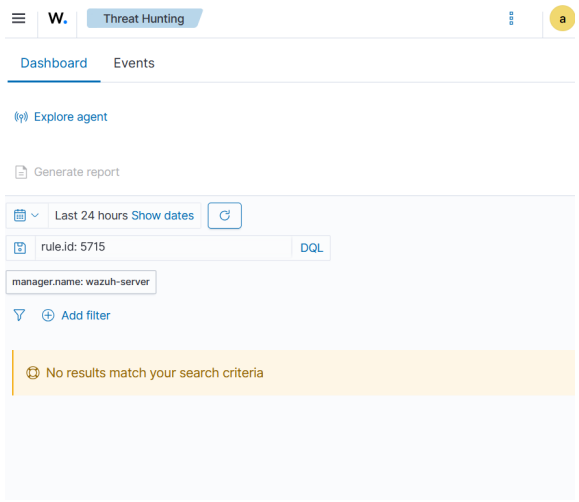


Figure 12: rule.id 5715 - no results match the search criteria.

A new file-integrity (FIM) alert was not captured during this Week 11 run (rules 550/553/554 returned no result in the last 24 hours). The FIM evidence from the Week 10 lab on 28 October 2026 is used instead: the test file /tmp/fim_test.txt was added (rule 554) and deleted (rule 553).

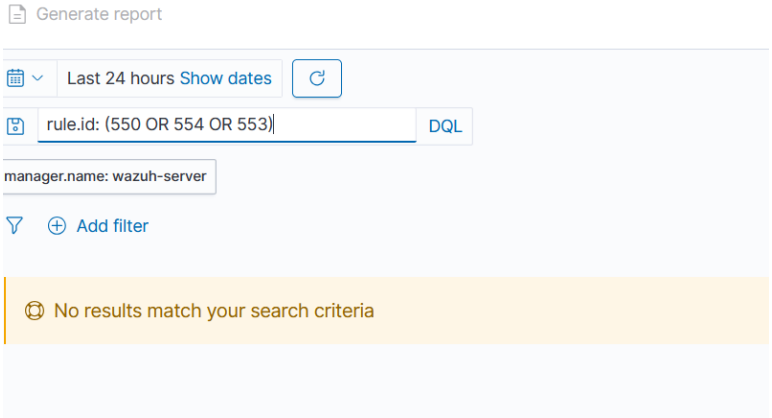
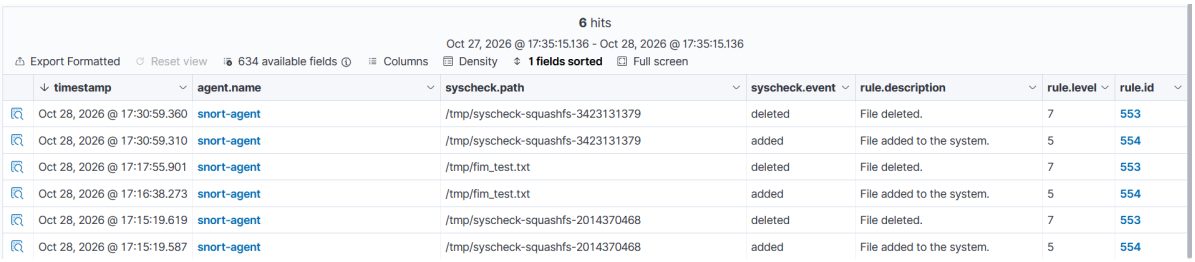


Figure 13: rule.id (550 OR 554 OR 553) in the last 24 hours - no results (no new file change in this run).



6 hits

Oct 27, 2026 @ 17:35:15.136 - Oct 28, 2026 @ 17:35:15.136

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timestamp	agent.name	syscheck.path	syscheck.event	rule.description	rule.level	rule.id
Oct 28, 2026 @ 17:30:59.360	snort-agent	/tmp/syscheck-squashfs-3423131379	deleted	File deleted.	7	553
Oct 28, 2026 @ 17:30:59.310	snort-agent	/tmp/syscheck-squashfs-3423131379	added	File added to the system.	5	554
Oct 28, 2026 @ 17:17:55.901	snort-agent	/tmp/fim_test.txt	deleted	File deleted.	7	553
Oct 28, 2026 @ 17:16:38.273	snort-agent	/tmp/fim_test.txt	added	File added to the system.	5	554
Oct 28, 2026 @ 17:15:19.619	snort-agent	/tmp/syscheck-squashfs-2014370468	deleted	File deleted.	7	553
Oct 28, 2026 @ 17:15:19.587	snort-agent	/tmp/syscheck-squashfs-2014370468	added	File added to the system.	5	554

Figure 14 (Week 10, 28 Oct 2026): FIM events - /tmp/fim_test.txt added (rule 554, level 5) and deleted (rule 553, level 7).

3.6 Security events collected (12 events + 1 correlation + context)

#	Time (PKT, Wazuh)	Category	Rule ID	Level	Description
1	14:06:24	Failed login	5710	5	sshd: Attempt to login using a non-existent user
2	14:06:38	Failed login	5710	5	sshd: Attempt to login using a non-existent user
3	14:06:50	Failed login	5710	5	sshd: Attempt to login using a non-existent user
4	14:06:54	Failed login	5710	5	sshd: Attempt to login using a non-existent user
5	14:06:56	Failed login	5710	5	sshd: Attempt to login using a non-existent user
6	14:07:02	Failed login	5710	5	sshd: Attempt to login using a non-existent user
7	14:07:07	Failed login	5710	5	sshd: Attempt to login using a non-existent user
8	14:07:17 (x2)	Failed login	5710	5	sshd: Attempt to login using a non-existent user (2 events same second)
9	14:07:30	Failed login (valid user)	5760	5	sshd: authentication failed (user snortadmin)
10	14:07:34	Failed login (valid user)	5760	5	sshd: authentication failed (user snortadmin)
11	(Wazuh dashboard)	Brute-force pattern	SSH brute-force class	10	One level-10 alert in the chart (Figure 8); rule ID not captured in a screenshot
12	14:10:36	Successful login after failures	40112	12	Multiple authentication failures followed by a success

#	Time (PKT, Wazuh)	Category	Rule ID	Level	Description
13	Agent status	Agent status	-	-	snort-agent Active (1), Disconnected (0)
14	Week 10 (28 Oct)	File change	554 / 553	5 / 7	File added / deleted: /tmp/fim_test.txt

3.7 Authentication, system events and security alerts - summary

Item to monitor	What was observed
Authentication events	17 authentication failures and 8 authentication successes in the last 24 hours (Figure 5).
Failed login attempts	Rules 5710 (non-existent user) and 5760 (authentication failed) from 192.168.56.1.
Successful login	snortadmin logged in over SSH at 09:10:59 UTC. Wazuh raised level-12 rule 40112.
File changes	None in this run. Week 10: /tmp/fim_test.txt added and deleted (rules 554, 553).
System events	VM clock change after resume (guest time sync event on the Wazuh VM console) and restart of wazuh-indexer, wazuh-manager and wazuh-dashboard.
Security alerts	0 critical (level 15+), 1 high (level 12+), several level 5 and 10 alerts.
Agent status	snort-agent Active.

4. Snort Monitoring

Snort was started on the host-only interface of soc-snort with this command (the first attempt failed because of a typo, `enp0s8` with a capital O instead of `enp0s8`):

```
sudo snort -A console -q -c /etc/snort/snort.conf -i enp0s8
```



Figure 15: First attempt failed with 'No such device exists' (typo in the interface name); the corrected command ran.

4.1 ICMP ping (Windows -> soc-snort)

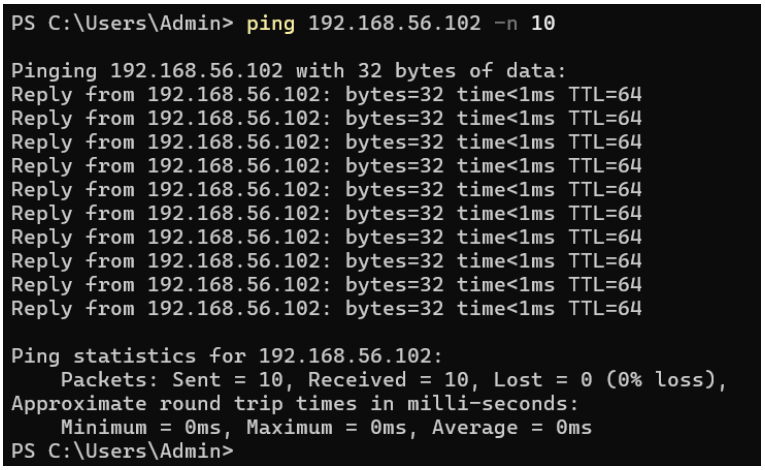


Figure 16: PowerShell ping 192.168.56.102 -n 10 - 10 sent, 10 received.

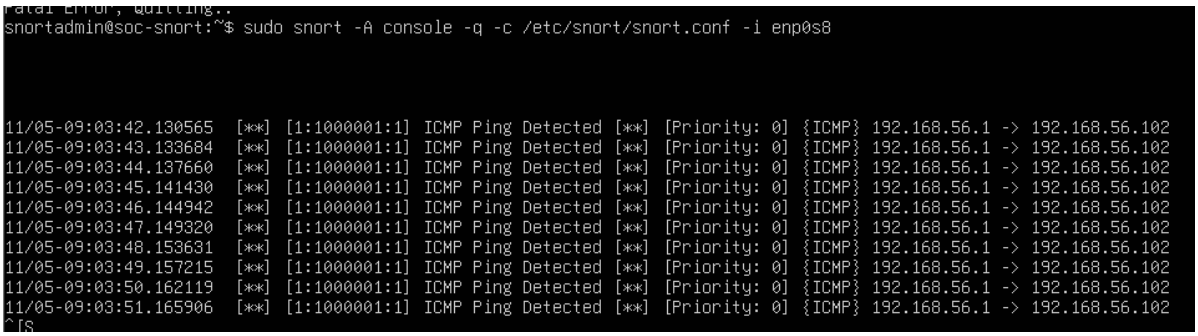


Figure 17: Snort alerts - 10 x [1:1000001:1] ICMP Ping Detected, 192.168.56.1 -> 192.168.56.102, 09:03:42 to 09:03:51 UTC.

4.2 Port scan (nmap SYN scan)

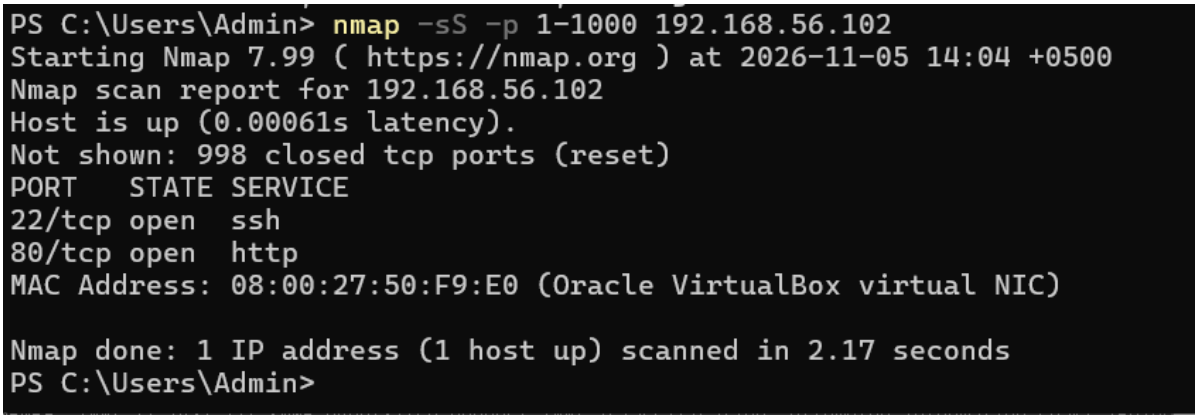


Figure 18: nmap -sS -p 1-1000 192.168.56.102 - ports 22/tcp (ssh) and 80/tcp (http) open, 998 closed.

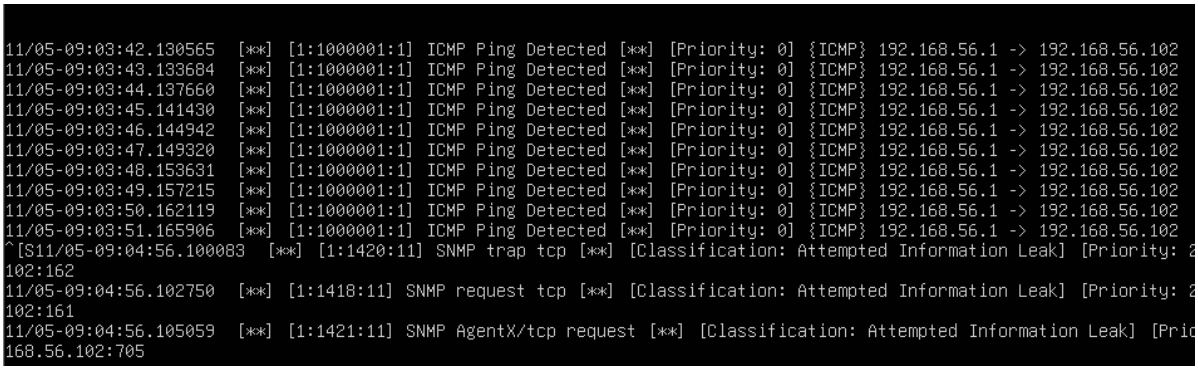


Figure 19: Snort alerts at 09:04:56 UTC: SNMP trap tcp (1:1420), SNMP request tcp (1:1418), SNMP AgentX/tcp request (1:1421).

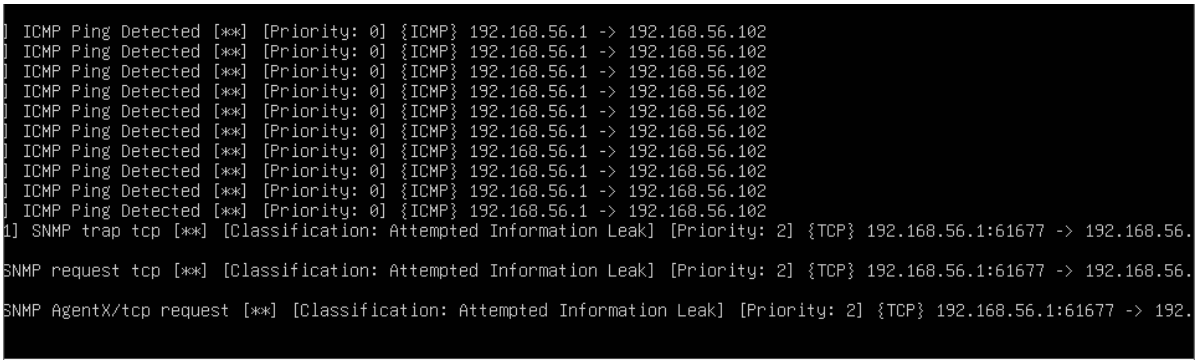


Figure 20: Same alerts (right side): Priority 2, Classification Attempted Information Leak, source 192.168.56.1:61677.

Explanation: my custom port-scan rule (SID 1000002) did not fire. The scan was detected **indirectly**: the SYN scan touched TCP ports 161, 162 and 705, and Snort's default SNMP signatures matched those connection attempts. This shows how a scan can show up under an unrelated rule name, and that a dedicated scan rule (with a threshold) should be tuned. No Snort alert for the SSH login attempts was observed in the captured output.

4.3 Snort alert summary

Time (UTC)	SID	Message	Priority	Source -> Destination
09:03:42 - 09:03:51	1:1000001	ICMP Ping Detected (10 alerts)	0	192.168.56.1 -> 192.168.56.102
09:04:56	1:1420	SNMP trap tcp	2	192.168.56.1:61677 -> 192.168.56.102:162
09:04:56	1:1418	SNMP request tcp	2	192.168.56.1:61677 -> 192.168.56.102:161

Time (UTC)	SID	Message	Priority	Source -> Destination
09:04:56	1:1421	SNMP AgentX/tcp request	2	192.168.56.1:61677 -> 192.168.56.102:705

5. Snort vs Wazuh Alert Comparison

Event	Snort Detection	Wazuh Detection	Severity	Analyst Finding
Ping (ICMP)	Yes - SID 1:1000001 (10 alerts)	No	Low	Normal activity from the admin host. No action, keep as baseline.
Port Scan (nmap SYN)	Yes - indirectly via SNMP SIDs 1:1418, 1:1420, 1:1421	No scan alert seen	Medium	Investigate. Source 192.168.56.1 is my own test host. Open ports 22 and 80 found.
Login Failure (SSH)	Not observed	Yes - rules 5710, 5760 (level 5)	Medium	Authentication event. 8 failures from one source IP in about 1 minute.
Brute-force pattern	Not observed	Yes - level 10 brute-force alert	Medium	Repeated failures from one IP confirmed by Wazuh.
Successful login after failures	Not observed (SSH is encrypted; no login signature)	Yes - rule 40112 (level 12)	High	Correlation: possible account compromise. Verify the user.
File change	No (network tool)	Yes in Week 10 (rules 554, 553). None in this run	Low-Medium	Expected for FIM test file in /tmp. Check user and process.
Agent status	No	Yes - Active	Informational	Monitoring healthy.

Why one tool detects an event and the other does not

- **Different data source.** Snort is a network IDS: it reads packets on the wire. Wazuh is a host-based SIEM/HIDS: it reads logs, file integrity data and agent events from inside the machine. Ping and port scans are network events, so Snort sees them; the Linux host logs very little about them, so Wazuh does not alert.
- **Different rules.** Snort alerts only if a signature matches. My ICMP rule matched; the port-scan rule SID 1000002 did not, but default SNMP signatures matched the scan's probes. Wazuh alerts only if a decoder and rule match a log line.
- **Encryption.** SSH is encrypted, so Snort cannot see a wrong password inside the session. The SSH server (sshd) writes the failure to auth.log, and Wazuh reads that log.
- **Context and correlation.** Wazuh can connect many log lines over time (rule 40112). Snort mostly evaluates single packets or short thresholds.
- **Conclusion:** the tools complement each other. Together they show the full story: Snort shows the scan from 192.168.56.1, Wazuh shows the login attempts from the same IP.

6. Alert Classification (16 alerts)

Severity scale used (aligned with the Wazuh dashboard): Informational = level 0-3, Low = 4-6, Medium = 7-11, High = 12-14, Critical = 15. Snort alerts are classified by priority and by context.

#	Alert / event	Source	Severity	Reason
1	ICMP Ping Detected (1:1000001)	Snort	Low	Ping is normal for diagnostics but is the first step of reconnaissance, so it is logged.
2	SNMP request tcp (1:1418)	Snort	Medium	Priority 2 information-leak signature; probing SNMP port 161.

#	Alert / event	Source	Severity	Reason
3	SNMP trap tcp (1:1420)	Snort	Medium	Same scan source; port 162 touched. Part of a reconnaissance pattern.
4	SNMP AgentX/tcp request (1:1421)	Snort	Medium	Port 705 touched by the same source; confirms wide port probing.
5	nmap SYN scan of ports 1-1000	Snort (indirect) / manual	Medium	Scanning finds open services; usually precedes an attack.
6	Unexpected open port 80/tcp (nmap result)	Manual finding	Low	An unused web service enlarges the attack surface; needs review.
7	Rule 5710 - login attempt using a non-existent user (x9)	Wazuh	Low	Level 5. A single one is common noise, but repeated attempts show username guessing.
8	Rule 5760 - sshd authentication failed (x2)	Wazuh	Low	Level 5. Valid username with wrong password; harmless alone.
9	SSH brute-force alert (level 10)	Wazuh	Medium	Many failures from one IP in a short time; maps to MITRE Brute Force.
10	Rule 40112 - multiple failures followed by a success	Wazuh	High	Level 12. Success after guessing is the strongest sign of account compromise.
11	SSH Accepted password for snortadmin from 192.168.56.1	auth.log	Medium	A valid login, but it follows failures from the same IP, so it needs verification.
12	File added: /tmp/fim_test.txt (rule 554)	Wazuh FIM (Week 10)	Low	Level 5. New file in /tmp; expected here, but unexpected files can be malware.
13	File deleted: /tmp/fim_test.txt (rule 553)	Wazuh FIM (Week 10)	Low	Level 7 in Wazuh, but deleting a temp test file is low risk; deletion can hide evidence in real cases.
14	sudo command run by snortadmin (auth.log)	auth.log	Informational	Normal admin activity (my own evidence commands). Only reviewed for privilege use.
15	Agent snort-agent status Active	Wazuh	Informational	Shows monitoring is healthy; no risk.
16	VM guest time change on Wazuh VM after resume	System event	Low	Clock drift (~22 s) can break log correlation and timelines, so it is tracked.

7. Incident Investigation

Scenario (safe, own lab): multiple failed SSH logins on a Linux system, followed by a successful login. The test was run on purpose from the Windows host (192.168.56.1) against soc-snort (192.168.56.102).

7.1 Evidence from the Linux server (auth.log)

```
PS C:\Users\Admin> ssh wronguser@192.168.56.102
wronguser@192.168.56.102's password:
Permission denied, please try again.
wronguser@192.168.56.102's password:
Permission denied, please try again.
wronguser@192.168.56.102's password:
wronguser@192.168.56.102: Permission denied (publickey,password).
PS C:\Users\Admin> ssh snortadmin@192.168.56.102
snortadmin@192.168.56.102's password:
Permission denied, please try again.
snortadmin@192.168.56.102's password:
Permission denied, please try again.
snortadmin@192.168.56.102's password:
```

Figure 21: PowerShell - repeated 'Permission denied' for wronguser and snortadmin.

```
snortadmin@soc-snort:~$ sudo grep -E "Failed password|Invalid user|Accepted password" /var/log/auth.log| tail -20
2026-10-28T12:02:03.771386+00:00 soc-snort sshd[5927]: Invalid user wronguser from 127.0.0.1 port 44314
2026-10-28T12:02:14.957309+00:00 soc-snort sshd[5927]: Failed password for invalid user wronguser from 127.0.0.1 port 4
2026-10-28T12:04:45.431563+00:00 soc-snort sshd[5942]: Invalid user wronguser from 192.168.56.1 port 34587
2026-10-28T12:04:49.591395+00:00 soc-snort sshd[5942]: Failed password for invalid user wronguser from 192.168.56.1 por
2026-10-28T12:04:54.022061+00:00 soc-snort sshd[5942]: Failed password for invalid user wronguser from 192.168.56.1 por
2026-11-05T09:06:46.641244+00:00 soc-snort sshd[2449]: Invalid user wronguser from 192.168.56.1 port 16144
2026-11-05T09:07:00.510020+00:00 soc-snort sshd[2449]: Failed password for invalid user wronguser from 192.168.56.1 por
2026-11-05T09:07:13.924232+00:00 soc-snort sshd[2449]: Failed password for invalid user wronguser from 192.168.56.1 por
2026-11-05T09:07:17.651139+00:00 soc-snort sshd[2449]: Failed password for invalid user wronguser from 192.168.56.1 por
2026-11-05T09:07:24.979134+00:00 soc-snort sshd[2451]: Invalid user wronguser from 192.168.56.1 port 51390
2026-11-05T09:07:31.107539+00:00 soc-snort sshd[2451]: Failed password for invalid user wronguser from 192.168.56.1 por
2026-11-05T09:07:36.700051+00:00 soc-snort sshd[2451]: Failed password for invalid user wronguser from 192.168.56.1 por
2026-11-05T09:07:40.045635+00:00 soc-snort sshd[2451]: Failed password for invalid user wronguser from 192.168.56.1 por
2026-11-05T09:07:52.213057+00:00 soc-snort sshd[2453]: Failed password for snortadmin from 192.168.56.1 port 39506 ssh2
2026-11-05T09:07:57.664280+00:00 soc-snort sshd[2453]: Failed password for snortadmin from 192.168.56.1 port 39506 ssh2
2026-11-05T09:09:52.540512+00:00 soc-snort sudo: snortadmin : TTY=tty1 ; PWD=/home/snortadmin ; USER=root ; COMMAND=/us
er|Accepted password' /var/log/auth.log
snortadmin@soc-snort:~$
```

Figure 22: auth.log - Invalid user wronguser and Failed password lines from 192.168.56.1 on 2026-11-05.

```
Connection closed by 192.168.56.102 port 22
PS C:\Users\Admin> ssh snortadmin@192.168.56.102
snortadmin@192.168.56.102's password:
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 6.8.0-147-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Nov  5 09:10:59 AM UTC 2026

  System load:             0.23
  Usage of /:              17.5% of 27.45GB
  Memory usage:            24%
  Swap usage:              0%
  Processes:               128
  Users logged in:         1
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe85:cefc

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.
```

Figure 23: Successful SSH login as snortadmin (Ubuntu 24.04.5 LTS banner, 09:10:59 UTC).

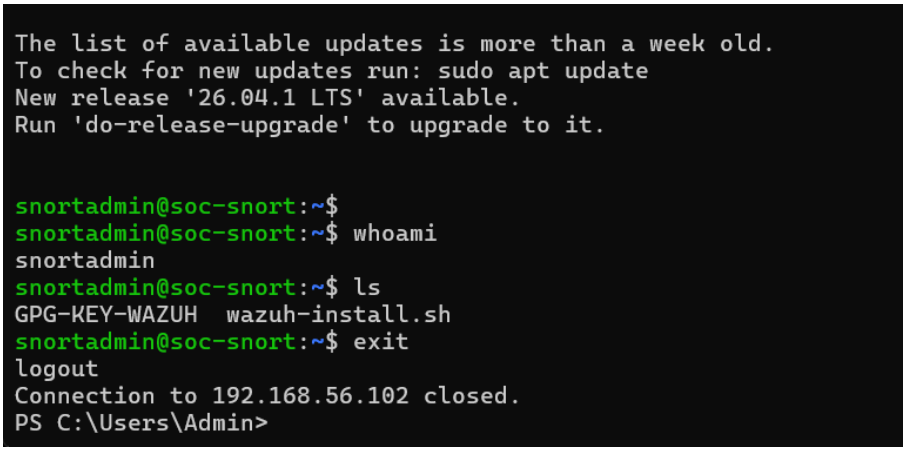


Figure 24: Commands after login: whoami, ls (GPG-KEY-WAZUH, wazuh-install.sh), exit.

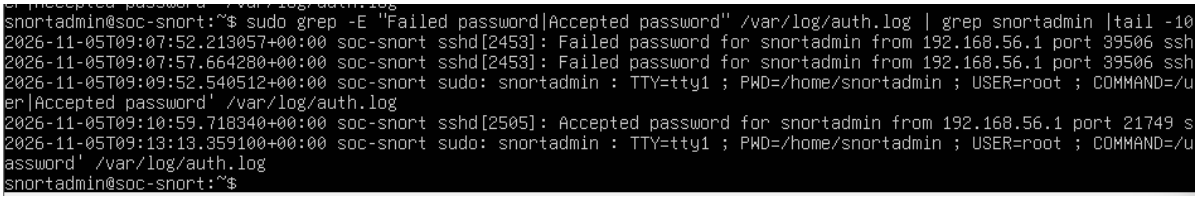


Figure 25: auth.log - Failed password for snortadmin at 09:07:52 and 09:07:57, then Accepted password at 09:10:59.

7.2 Timeline (UTC; PKT = UTC+5)

Time (UTC)	Event	Source
09:03:42 - 09:03:51	10 ICMP pings from 192.168.56.1	Snort 1:1000001
09:04:56	nmap SYN scan; SNMP signatures fire	Snort 1:1418, 1:1420, 1:1421
09:06:46	**First** failed activity: Invalid user wronguser (SSH session PID 2449)	auth.log
09:07:00 - 09:07:17	Failed password for invalid user wronguser (session 2449)	auth.log / Wazuh 5710
09:07:24 - 09:07:40	Second session (PID 2451): invalid user wronguser, 3 more failures	auth.log / Wazuh 5710
09:07:52, 09:07:57	Failed password for **snortadmin** (valid account), PID 2453	auth.log / Wazuh 5760
09:10:59	**Accepted password for snortadmin** from 192.168.56.1 port 21749 (PID 2505)	auth.log
~09:11	Post-login commands: whoami, ls, exit (no privilege escalation seen)	SSH session
14:10:36 PKT (Wazuh clock)	Level 12 alert: multiple authentication failures followed by a success	Wazuh 40112

7.3 Investigation questions

Question	Answer
When did the activity start?	5 Nov 2026, 09:06:46 UTC (14:06:46 PKT) - first invalid-user SSH attempt. Network recon began earlier at 09:03:42 UTC.
Which account was targeted?	**wronguser** (does not exist) and **snortadmin** (valid account).
Source IP?	192.168.56.1 (Windows host of the lab). Target: 192.168.56.102.
Number of failed attempts?	8 failed-password events in auth.log for this incident: 6 for wronguser and 2 for snortadmin. Wazuh recorded 11 level-5 alerts (9 x rule 5710, 2 x rule 5760).
Was there a successful login?	Yes. snortadmin, Accepted password, 09:10:59 UTC, port 21749.

Question	Answer
What happened after login?	whoami, ls and exit within one short session. Files visible: GPG-KEY-WAZUH and wazuh-install.sh. No file change, no new user and no sudo from this SSH session were seen.
Which Wazuh rule generated the alert?	5710 and 5760 (level 5), a level-10 brute-force alert, and correlation rule **40112** (level 12).
Did Snort detect related network activity?	Yes. Same source IP: ICMP ping (SID 1:1000001) and the port scan through SNMP SIDs 1:1418, 1:1420, 1:1421. No Snort alert for the SSH logins was observed.
True Positive or False Positive?	**True Positive.** The detection logic was correct: the pattern is a real brute-force-then-login sequence. The activity itself was an authorized simulation I ran in my own lab. In a production network this would be treated as a suspected account compromise.

7.4 Analyst conclusion

The same IP first pinged and scanned the server, then guessed a non-existent username, then guessed the password of a real account, and then logged in. Wazuh and Snort together reconstruct this chain. Severity is set to **High** because of the success after repeated failures (rule 40112, level 12).

8. IOC Identification

IOC type	Value	Why it matters
IP address (source)	192.168.56.1	Origin of the ping, scan and SSH attempts. Block or investigate; search other logs for this IP.
IP address (target)	192.168.56.102 (soc-snort)	Affected host; helps scope impact.
Username (invalid)	wronguser	Non-existent account name = username guessing. Good detection keyword.
Username (targeted)	snortadmin	Valid account that was attacked and then logged in. Reset credentials and review.
File name / path	/var/log/auth.log	Main evidence source for SSH failures and success.
File name / path	/home/snortadmin (GPG-KEY-WAZUH, wazuh-install.sh)	Files seen by the session after login; check they were not read or copied.
File name / path (Week 10 FIM)	/tmp/fim_test.txt	File added (554) and deleted (553); shows FIM works. Unexpected files in /tmp can indicate malware.
Domain	None observed	All traffic was local (host-only). In a real case, check DNS logs for C2 domains. Hostnames: soc-snort, wazuh-server.
Port	22/tcp (SSH)	Service attacked. Restrict by firewall and use keys.
Port	80/tcp (HTTP)	Open and unexpected; review or close.
Port	161, 162, 705 (SNMP / AgentX)	Probed by the scan and matched by Snort SIDs 1418, 1420, 1421.
Port (source)	61677 (scan), 21749 (SSH session)	Ephemeral source ports link a Snort alert to a specific connection.
Protocol	ICMP, TCP, SSH	ICMP = ping/recon; TCP SYN = scan; SSH = target of brute force.
Timestamp	2026-11-05 09:03:42 - 09:10:59 UTC	Defines the incident window for log searches and the timeline.
Wazuh Rule ID	5710, 5760, level-10 brute force, 40112	Link alerts to documented detections and MITRE techniques (Brute Force, Valid Accounts).
Snort SID	1:1000001, 1:1418, 1:1420, 1:1421	Identify which signature matched; used for tuning and blocking rules.
MAC (target NIC)	08:00:27:50:F9:E0	Hardware address seen in the nmap result; confirms the host in the local network.

9. Incident Response Procedure

Incident: **DG-SOC-001 - Multiple failed SSH logins followed by a successful login on soc-snort.**

Phase	Actions for this incident
1. Preparation	SOC should already have: Wazuh agents on all servers, Snort/IDS on key network segments, central log collection with correct time sync (NTP), tuned rules (including SSH brute-force and port-scan rules), a ticketing system, playbooks, contact list and escalation path, backups and a clean admin workstation. Staff trained on L1/L2/L3 duties.
2. Identification	Wazuh raised rules 5710, 5760, a level-10 brute-force alert, then rule 40112 (level 12). Snort showed ping and scan from the same IP. L1 opened ticket DG-SOC-001 and escalated to L2 because of the login after failures.

Phase	Actions for this incident
3. Analysis	Evidence collected: auth.log lines (failed and accepted), Wazuh alerts with timestamps, Snort alerts (SIDs, source IP/ports), PowerShell/SSH session screenshots, list of files in the home folder. Confirmed source 192.168.56.1, accounts wronguser and snortadmin, 8 failures, 1 success, no privilege escalation.
4. Containment	In production: block 192.168.56.1 at the firewall (for example `sudo ufw deny from 192.168.56.1`), terminate the SSH session, lock or force a password reset on snortadmin, and isolate the host if suspicious commands are found. In this lab: no block was needed because the source is my own test machine.
5. Eradication	Remove the root cause: change the snortadmin password, check ~/.ssh/authorized_keys and cron jobs for persistence, review new users and sudo changes, disable password login for SSH (use keys), install pending updates (16 updates were available), and close unused port 80.
6. Recovery	Restore normal access with the new credentials, confirm services work, keep enhanced monitoring on snortadmin and the SSH service for several days, and make sure Wazuh and Snort are healthy and clocks are synchronized.
7. Lessons Learned	Enable fail2ban or SSH rate limiting, enforce MFA/keys, tune Snort SSH brute-force and port-scan rules, reduce exposed services, keep VM clocks synced, keep FIM real-time on important folders, and update the playbooks. Write a short post-incident review.

10. SOC Incident Ticket

Field	Details
Incident ID	DG-SOC-001
Title	Multiple failed SSH logins followed by a successful login on soc-snort
Date/Time	2026-11-05, 09:06 - 09:11 UTC (14:06 - 14:11 PKT)
Severity	High (Wazuh rule 40112, level 12)
Affected System	soc-snort, Ubuntu 24.04 LTS, 192.168.56.102 (Wazuh agent name: snort-agent)
Source IP	192.168.56.1
User/Account	wronguser (non-existent), snortadmin (valid, later logged in)
Detection Source	Wazuh and Snort
Description	A single source IP pinged and scanned the server, then made repeated SSH login attempts using a non-existent user and a valid user. After the failures, snortadmin logged in successfully. Wazuh correlated the failures and the success into a level-12 alert.
Evidence	auth.log: 8 Failed password events (09:07:00 - 09:07:57 UTC) and Accepted password at 09:10:59 UTC. Wazuh: 11 x level-5 alerts (5710, 5760), 1 x level-10 alert, 1 x rule 40112. Snort: 1:1000001 (10 alerts), 1:1418, 1:1420, 1:1421. Screenshots: Figures 5-12 and 16-25.
IOC	IP 192.168.56.1; users wronguser, snortadmin; ports 22, 80, 161, 162, 705; protocols ICMP, TCP, SSH; Wazuh rules 5710, 5760, 40112; Snort SIDs 1:1000001, 1:1418, 1:1420, 1:1421.
Investigation	Timeline built from auth.log, Wazuh and Snort. Source is the analyst's own Windows host. After login only whoami, ls and exit were run. No privilege escalation, no new files, no persistence found. Verdict: True Positive detection of an authorized simulation.
Action Taken	Evidence collected and documented. Lab source confirmed authorized, so no block was applied. Recommended production actions listed below.
Current Status	Resolved (authorized lab simulation). In production this would stay Investigating until the user is verified.
Recommendation	Block or rate-limit the source IP; reset snortadmin password and review the account; disable SSH password login and use keys; install fail2ban; add a Snort SSH brute-force rule and a tuned port-scan rule; close unused port 80; apply the pending updates; keep FIM real-time on key directories; synchronize VM clocks.

11. Alert Response Playbooks

Playbook 1 - Multiple Failed Logins

Step	Actions
Detection	Wazuh rules 5710, 5760, 5712 (brute force) or 40112. Alert threshold: 5 or more failures from one IP in 60 seconds.
Investigation	Find source IP, targeted usernames, number and time range of failures: `grep 'Failed password' /var/log/auth.log`. Check if the user exists and if the IP is known.
Verification	Was there a success after the failures? `grep 'Accepted' /var/log/auth.log` or Wazuh rule 40112. Ask the user if it was them. Check geolocation / asset list. Decide True or False Positive.
Containment	Block the IP, lock the account or force password reset, kill active sessions (`pkill -u user`), enable fail2ban.
Escalation	Escalate to L2 if a success followed the failures, a privileged account was targeted, or many accounts were tried. Escalate to L3/IR lead if post-login activity is suspicious.

Playbook 2 - Port Scan

Step	Actions
Detection	Snort scan or related signatures (for example SIDs 1:1418, 1:1420, 1:1421, or a threshold rule such as SID 1000002), or many connection attempts from one IP.
Identify source	Get source IP and ports from the alert (for example 192.168.56.1:61677). Count ports and the time window. Check whether the IP is internal or external.
Verify authorization	Check with the vulnerability-management or IT team whether a scheduled scan or an approved pentest is running. Compare with asset and scan-schedule lists.
Investigate	Search for the same IP in Wazuh and firewall logs: later logins, exploit attempts, connections to open ports (22, 80). Review which ports were open.
Response	If authorized: close the alert and document it. If unauthorized: block the IP, close unneeded ports, increase monitoring on the exposed services, and escalate if follow-up attacks are seen.

Playbook 3 - Suspicious File / System Change

Step	Actions
Detection	Wazuh FIM rules 550 (modified), 553 (deleted), 554 (added), or changes in /etc, /bin, /usr, startup folders.
Identify file	Read syscheck.path, event type, size, permissions and hashes in the alert. Note the time. Decide how important the file is (config, binary, temp file).
Check user / process	Which user changed it (who-data audit or auth logs)? Which process? Was it during a change window or a patch? `last -a`, `ps aux`, `ls -l`.
Investigate	Compare the file hash with a known-good value or VirusTotal, check for new users, cron jobs, SSH keys and startup entries, and check for related login alerts.
Response	If authorized: document and close. If malicious: isolate the host, restore the file from backup, remove persistence, reset credentials, and escalate to L2/L3.

12. Findings and Recommendations

Findings

- Snort detected the ping (SID 1:1000001) and the nmap scan, but the scan was detected only **indirectly** through default SNMP signatures. My custom port-scan rule (SID 1000002) did not fire.
- Wazuh detected the failed SSH logins (rules 5710, 5760), a brute-force pattern (level 10) and the login after failures (rule 40112, level 12), mapped to MITRE Brute Force and Valid Accounts.
- No Snort alert for the SSH attempts was seen, because SSH is encrypted and no SSH rule was matched.
- Rule 5715 produced no result; the correlation rule 40112 covered the successful login.
- A new file-change alert was not captured in this run; the Week 10 FIM alerts (rules 554, 553 on /tmp/fim_test.txt) show that FIM works.
- nmap found unexpected open port 80/tcp besides SSH (22/tcp).
- Wazuh timestamps were about 22-23 seconds behind the Linux logs after the Wazuh VM was resumed.
- Ubuntu showed 16 pending updates and a new release available.

Recommendations

- **Harden SSH:** key-based login, disable password authentication, disable root login, limit allowed users, change or restrict the port, add fail2ban.
- **Tune Snort:** add a threshold-based port-scan rule and an SSH brute-force rule (for example SID 1000003) and test that they fire; check that HOME_NET is set to 192.168.56.0/24.
- **Tune Wazuh:** enable real-time FIM on important folders, add active response to block repeated offenders, and create dashboards for rules 5710, 5760, 40112.
- **Reduce attack surface:** close or secure port 80 if not needed; disable unused SNMP services; apply pending updates.
- **Time sync:** use NTP on all VMs so logs from Snort, Wazuh and Linux line up.
- **Process:** keep tickets, playbooks and a regular review of alerts; train L1 analysts on correlation.

13. Conclusion

This lab showed how a SOC analyst combines network alerts from Snort and host alerts from Wazuh to understand one incident. Snort showed the reconnaissance (ping and scan) and Wazuh showed the authentication attacks and the correlated login. Neither tool alone told the full story. By building a timeline, extracting IOCs, classifying 16 alerts, applying the incident response steps, and writing a ticket and three playbooks, I practiced the full cycle from detection to lessons learned. The incident (DG-SOC-001) is a True Positive detection of an authorized simulation. Its main lessons are to harden SSH, tune the Snort rules, keep FIM and time sync healthy, and always correlate alerts before deciding severity.